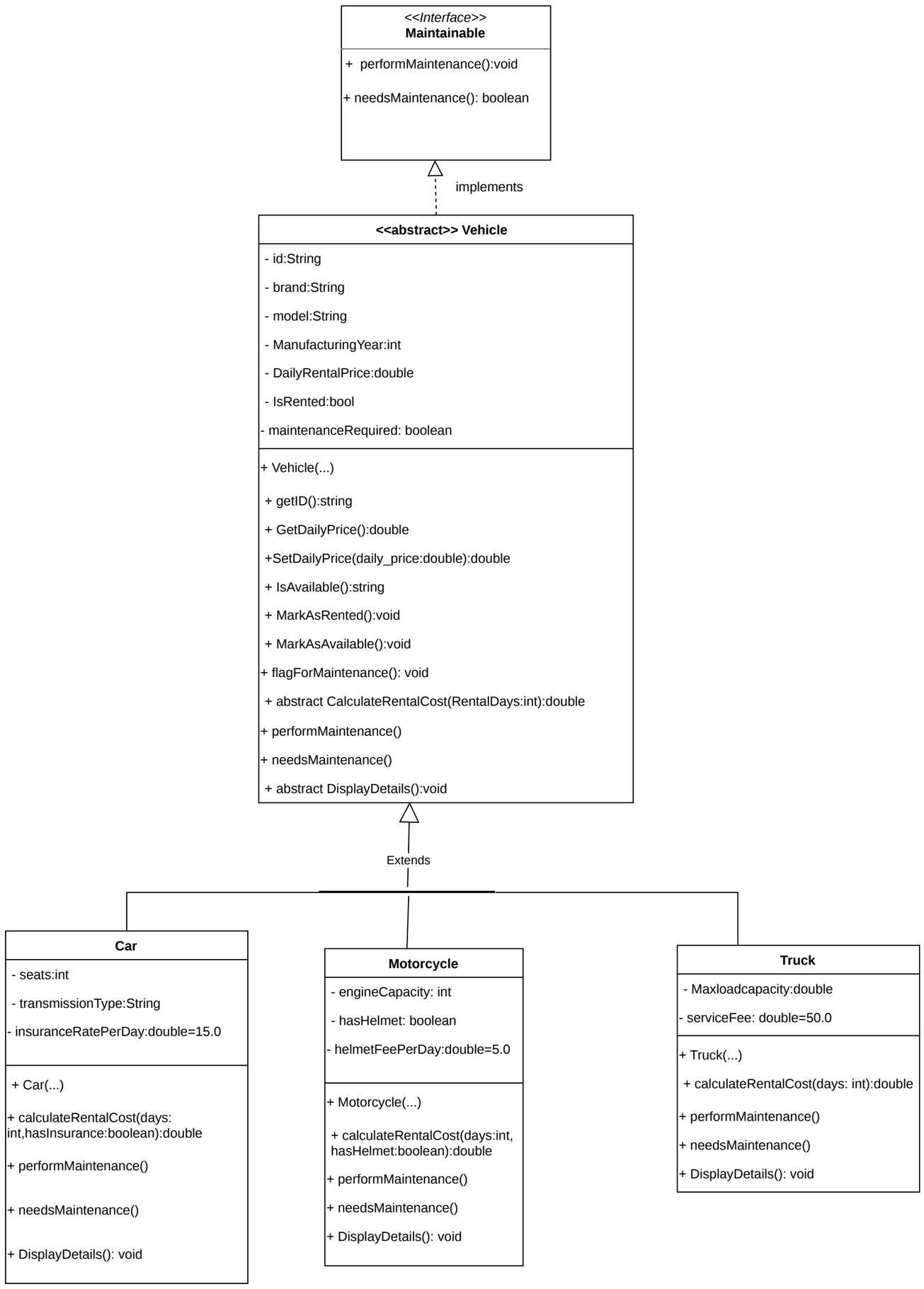
<<Interface>>
Maintainable
+ performMaintenance():void
+ needsMaintenance(): boolean
implements
<<abstract>> Vehicle
- id:String
- brand:String
- model:String
- ManufacturingYear:int
- DailyRentalPrice:double
- IsRented:bool
- maintenanceRequired: boolean
+ Vehicle(...)
+ getID():string
+ GetDailyPrice():double
+SetDailyPrice(daily_price:double):double
+ IsAvailable():string
+ MarkAsRented():void
+ MarkAsAvailable():void
+ flagForMaintenance(): void
+ abstract CalculateRentalCost(RentalDays:int):double
+ performMaintenance()
+ needsMaintenance()
+ abstract DisplayDetails():void
Extends
Car
- seats:int
- transmissionType:String
- insuranceRatePerDay:double=15.0
+ Car(...)
+ calculateRentalCost(days: int,hasInsurance:boolean):double
+ performMaintenance()
+ needsMaintenance()
+ DisplayDetails(): void
Motorcycle
- engineCapacity: int
- hasHelmet: boolean
- helmetFeePerDay:double=5.0
+ Motorcycle(...)
+ calculateRentalCost(days:int, hasHelmet:boolean):double
+ performMaintenance()
+ needsMaintenance()
+ DisplayDetails(): void
Truck
- Maxloadcapacity:double
- serviceFee: double=50.0
+ Truck(...)
+ calculateRentalCost(days: int):double
+ performMaintenance()
+ needsMaintenance()
+ DisplayDetails(): void

RentalManagement

- rentalsList: List<RentalTransaction>
- customerManager: ArrayList<Customer>
- vehicleManager: ArrayList<Vehicle>

+ processRental(rentalId:String, customerId:String, vehicleId:String, days:int): boolean
+ calculateTotalCost(vehicle: Vehicle, days: int): double
+ processReturn(rentalId: String): boolean
+ listAllRentals(): void
+ listActiveRentals(): void
+ calculateTotalCompletedRevenue(): double

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0....*

RentalTransaction

- transId:String
- customer: Customer
- vehicle: Vehicle
- rentalDays: int
- totalCost: double
- status: String

+ RentalTransaction(rentalId: String, customer: Customer, vehicle: Vehicle,days:int,totalcost:double)
+ getRentalId(): String
+ getCustomer(): Customer
+ getVehicle(): Vehicle
+ getRentalDays(): int
+ getTotalCost(): double
+ getStatus(): String
+ completeRental(): void
+ displayTransactionDetails(): void

reference to

Its implementation exists in the previous page

Vehicle

Customer

- Cus_id:String
- fullName:String
- phone:String
- Email:String
- DLnumber:String

+ Customer(id: String, name: String, phone: String, email: String, dlNumber: String)
+ getCustomerId():String
+ getFullName():String
+ getPhone():String
+ getEmail():String
+ getDlNumber():String
+ setPhoneNumber(newPhone: String):void
+ setEmailAddress(newEmail:String):void
+ getDlNumber(): String
+ displayCustomerDetails():void